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UAMFDet: Acoustic-Optical Fusion for Underwater Multi-Modal Object Detection

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ABSTRACT

Underwater object detection serves as a crucial means for autonomous underwater vehicles (AUVs) to gain awareness of their surroundings. Currently, AUVs predominantly depend on underwater optical cameras or sonar sensing techniques to furnish vital information sources for subsequent tasks such as underwater rescue and mining exploration. However, the influence of underwater light attenuation or significant background noise often leads to the failure of either the acoustic or optical sensor. Consequently, the traditional single-modal object detection network, which relies exclusively on either the optical or acoustic modality, struggles to adapt to the varying complexities of underwater environments. To address this challenge, this paper proposes a novel underwater acoustic-optical fusion-based underwater multi-modal object detection paradigm termed UAMFDet, which fuses highly misaligned acoustic-optical features in the spatial dimension at both the fine-grained level and the instance level. First, we propose a multi-modal deformable self-aligned feature fusion module to adaptively capture feature dependencies between multi-modal targets, and perform self-aligned multi-modal fine-grained feature fusion by differential fusion. Then a multi-modal instance-level feature matching network is designed. It matches multi-modal instance features by a lightweight cross-attention mechanism and performs differential fusion to achieve instance-level feature fusion. In addition, we establish a data set dedicated to underwater acoustic-optical fusion object detection tasks called UAOF, and conduct a large number of experiments on the UAOF data set to verify the effectiveness of UAMFDet.

1 | Introduction

As land resources become increasingly scarce, the exploration of marine environments has emerged as a new hot spot (Raveendran, Patil, and Birajdar 2021; Xu et al. 2023). Autonomous underwater vehicles (AUVs), equipped with a diverse array of sensors for underwater environmental perception, represent a formidable tool for autonomous underwater operations. As the basis of AUV's cognition of the underwater environment, underwater object detection technology is a prerequisite for AUVs to carry out autonomous underwater exploration (Fayaz et al. 2024). For instance, a typical application scenario is an underwater emergency rescue, where an AUV needs to accurately locate the position of the drowning person for subsequent rescue tasks. Currently, AUVs mainly rely on sonar or optical cameras for underwater sensing and carry out underwater object detection tasks based on the captured images. However, the two types of sensors often encounter significant limitations in underwater imaging. While optical images represent richer target features, their ability to capture long-distance imagery is hampered by the rapid attenuation of optical media underwater (Zhao et al. 2021).

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In addition, the varying attenuation degrees of different wavelengths lead to a pronounced blue-green bias in optical images. Conversely, sonar can achieve long-distance imaging but fails to capture nuanced target characteristics such as color and texture, as its imaging relies solely on echo intensity (Yu et al. 2021). These challenges significantly constrain the performance and advancement of subsequent underwater target detection technology. Integrating the advantages of acoustic and optical models is one of the effective ways to solve these problems (Remmas, Chemori, and Kruusmaa 2021).

In recent years, multi-modal fusion technology has developed rapidly in the field of autonomous driving (Zhang, Gong, et al. 2023). By fusing features from diverse modalities, a comprehensive and robust target representation is achieved, which can adeptly adapt to a wide range of perception scenarios. For instance, during nighttime, when camera visibility is compromised, infrared sensor imaging effectively mitigates the loss of visible light information. Inspired by this, we endeavor to explore a novel underwater multi-modal object detection paradigm. It aims to fuse optical modality with rich feature information and acoustic modality with accurate location information to achieve modal complementarity in different detection scenarios.

The predominant multi-modal fusion approach involves the design of intricate local feature fusion techniques, which rely on calibrating and aligning images from different modalities (Zhang, Liu, et al. 2021; Tu et al. 2022; Chaoxia et al. 2023). However, as illustrated in Figure 1, acoustic and optical images exhibit significantly distinct imaging dimensions. Specifically, optical images are captured in the X–Y plane of the image coordination system, whereas acoustic images are represented in the X–Z plane based on sonar echo intensity. This inherent discrepancy poses challenges in deriving an accurate calibration matrix for spatially aligning images from these two modalities, rendering most terrestrial multi-modal fusion methods inapplicable in this context.

To address the aforementioned challenges, we introduce a novel underwater multi-modal object detection network for deployment on AUVs, called UAMFDet. In UAMFDet, we eliminate the costly inter-modality calibration step and, instead, adaptively capture the alignment relationship between acoustic and optical modalities. Our approach involves fusing features through fine-grained feature-level and instance-level fusion. First, we propose a multi-modal deformable self-aligned feature fusion (MDSFF) module. It leverages a deformable attention mechanism to establish feature dependencies between modalities, thereby achieving automatic alignment of multi-modal features. The self-aligned multi-modal features are fused through a differential fusion module to enhance feature complementarity across modalities. Unlike intensive prediction tasks like semantic segmentation, object detection needs to predict the instance object by box regression, so it also pays attention to the feature representation at the instance level. We

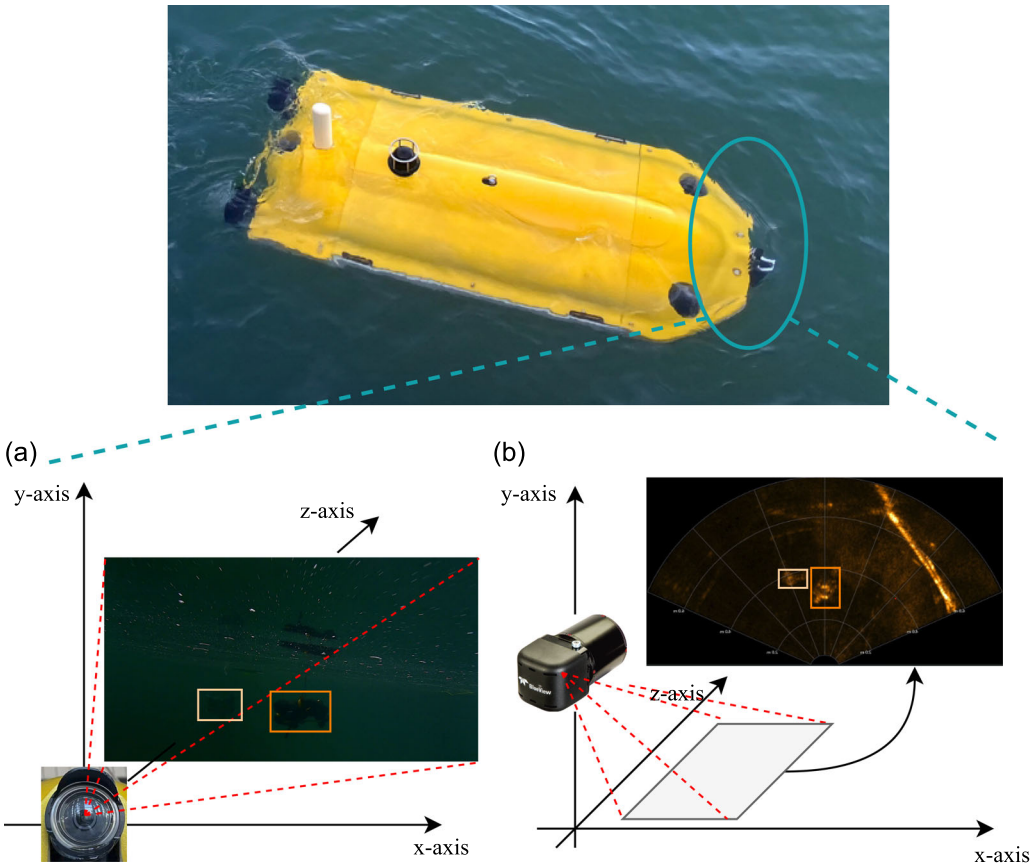


FIGURE 1 | Imaging dimensions of underwater optical cameras and forward-looking sonar for AUVs.

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further introduce a multi-modal instance-level feature matching (MIFM) module. It extracts region proposals for different modalities using a parameter-shared region proposal network (RPN) and then matches the multi-modal instance features of the proposal region through a lightweight cross-attention mechanism, which allows for instance-level feature fusion. By employing these two fusion processes at different levels, UAMFDet effectively integrates features from acoustic and optical modalities. UAMFDet differs from other related work in that our work omits the physical registration process and focuses on adaptive alignment and effective complementation of features derived from two distinct acoustic-optic modalities, while other work is based on paired images registered through projection matrices for multi-modal local feature fusion. Moreover, as there is no publicly available benchmark for acoustic-optical fusion object detection, we propose the underwater acoustic-optical fusion object detection data set (UAOF). To the best of our knowledge, UAMFDet is the first underwater acoustic-optical fusion object detector that performs multi-modal fusion in the feature fusion stage and UAOF is the first data set in the field of underwater acoustic-optical fusion object detection. The numerous results obtained on the UAOF data set validate the effectiveness of our proposed method.

Our contribution can be summarized as follows:

(1) We explore a novel underwater multi-modal fusion task for AUVs and propose the underwater acoustic-optical multi-modal fusion object detector (UAMFDet). In UAMFDet, an MDSFF module is proposed to align and fuse the fine-grained features of acoustic and optical models. Furthermore, a MIFM module is introduced that matches acoustic and optical instance features, enabling instance-level feature fusion.

(2) For the underwater acoustic-optical fusion object detection task, we construct an underwater acoustic-optical fusion object detection data set UAOF. To the best of our knowledge, this is the first data set for the domain of underwater acoustic-optical object detection.

(3) We verify the effectiveness of the proposed UAMFDet on the UAOF data set. The numerous results obtained on the UAOF data set validate the effectiveness of our proposed method on different underwater missions.

2 | Related Works

2.1 | Object Detection

As a fundamental task in computer vision, object detection aims to classify objects and predict their bounding boxes concurrently. With the application of deep learning technology, the field of object detection has developed rapidly (Liu et al. 2020). Currently, the mainstream object detection networks are broadly categorized into two types: one-stage and two-stage object detection networks. The one-stage object detection network directly predicts the class and box regression parameters of the target through object detection heads. The YOLO algorithm (Redmon and Farhadi 2018; Wang, Bochkovskiy, and Liao 2021, 2023; Ge et al. 2021) divides the feature map into uniform grid intervals, allowing the classification and box regression parameters of a target to be directly predicted by the grid cell whose center coincides with the target's position. RetinaNet addresses the imbalance in object detection tasks, where positive samples are scarce compared to negative ones, by introducing Focal loss to balance the ratio between the two types (Lin et al. 2017). The two-stage object detection network follows a different approach. It first generates candidate boxes for the target through an intermediate network and then fine-tunes these boxes to achieve more accurate detection boxes. Representative algorithms in this category are the Faster-RCNN series (Girshick 2015; Ren et al. 2017), Cascade-RCNN (Cai and Vasconcelos 2018), and Dynamic-RCNN (Zhang et al. 2020). Faster-RCNN generates candidate proposals via the RPN network and then refines them through the fully connected detection head to obtain the final detection box. Cascade-RCNN leverages the observation that the output of one detector serves as a good distribution for training the next higher-quality detector, achieving this by cascading a series of detectors with increasing IoU thresholds. Dynamic-RCNN, on the other hand, automatically adapts the label assignment criteria and the shape of the regression loss function to the specific data set, based on training statistics. All of the aforementioned object detection networks rely on convolutional neural networks for feature learning. With the development of transformer in computer vision, DeTR series object detection algorithms have gained popularity (Carion et al. 2020; Dai, Chen, et al. 2021; Dai, Cai, et al., 2021; Meng et al. 2021). These algorithms forego the need for NMS post-processing and directly predict the object category and detection box in an end-to-end manner through Hungarian matching.

Despite the remarkable progress in the field of general object detection algorithms, their efficacy when applied to underwater environments is significantly hindered. This limitation stems from the severe image degradation that occurs as a result of the unique and complex characteristics of the underwater imaging environment. These conditions introduce a set of challenges that are not typically encountered in terrestrial scenarios, necessitating the development of specialized detection technologies that can effectively navigate the intricacies of underwater perception.

2.2 | Underwater Object Detection

Due to the intricate underwater imaging conditions, underwater object detection networks progress comparatively slower compared with land object detectors (Song, He, and Liu 2021). Categorizing the imaging sensors employed, the underwater object detection network can be divided into underwater optical and acoustic object detection. In the field of underwater optical object detection, researchers are primarily preoccupied with addressing the hindrance of color cast and severe blur on target detection accuracy. For instance, Ye et al. jointly trained a color conversion network and an object detection network to mitigate the impact of high turquoise bias on object detection (Yeh et al. 2022). SWIPENet introduces an inverse multi-scale adaptive enhancement algorithm to minimize the effects of noise in underwater images (Chen et al. 2022). Boosting-RCNN

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captures underwater fuzzy targets with uncertainty modeling and hard power mining angle (Song et al. 2023).

In the domain of underwater acoustic-based object detection, researchers are primarily concerned with enhancing the impact of poor target features on target detection in sonar images. MLFFNet leverages a multi-scale convolution module to extract multi-scale features of the target in sonar images, thereby enhancing the target's feature representation (Wang et al. 2022). TransYOLO establishes the global dependency of target features by integrating the transformer and introduces a novel ellipse quality evaluation method (EQE) to improve the reliability of detection box localization quality estimation (Yuanzi, Xiufen, and Weizheng 2022).

While object detection algorithms grounded in a single modality have indeed made notable advancements, they still face limitations when deployed in a variety of complex scenarios. To attain a more reliable form of environmental perception, researchers have begun to explore more robust multimodal perception tasks.

2.3 | Multi-Modal Feature Fusion

Despite significant advances in visible light-based perception tasks in recent years, its application in complex scenarios like nighttime low-light environments, adverse weather conditions, and heavily occluded scenes remains challenging due to imaging constraints. Researchers have proposed to supplement the visible light modality with information from different modalities to obtain more robust detection results. In the realm of general target perception, radar point cloud information, depth information, and infrared data are commonly employed to complement visible light. Radar point cloud provides more three-dimensional (3D) spatial information about the target, which is mainly applied in the field of 3D target detection (Liu et al. 2023; Li et al. 2022; Cai et al. 2023). Depth maps are used to supplement object features in indoor scenes by providing more robust geometric shapes. Representative algorithms include SA-GATE (Chen et al. 2020), ACNet (Hu et al. 2019), and Depth-aware CNN (Wang and Neumann 2018). Infrared images can provide multi-spectral information in dark scenes and supplement visible light because of the feature information lost in low-light conditions. The representative algorithms are ABMDRNet (Zhang, Zhao, et al. 2021), GMNet (Zhou et al. 2021), EAEFNet (Liang et al. 2023), and so forth.

All the aforementioned methods achieve the complementarity of multi-modal local features through intricate feature fusion modules. However, these approaches rely on the assumption that images from diverse modalities are precisely aligned in spatial dimensions. Currently, the realm of weak-aligned multimodal target detection remains relatively understudied. Zhang, Liu, et al. (2021) have devised an innovative region proposal frame jitter technique, effectively fusing weak-aligned features from optical and infrared modalities. In addition, Chaoxia et al. (2023) employed a projection methodology to ascertain the approximate flame location in the optical mode within thermal imagery, enhancing flame detection capabilities. Tu et al. (2022), on the other hand, introduced a modality alignment module leveraging eigenby-eigenaffine transformation and dynamic convolution, thereby meticulously modeling the profound intermodal correlations. Contrary to visible-radar or visible-infrared modalities, aligning underwater acoustic and optical images through calibration is particularly challenging. Therefore, enabling the network to adaptively capture the intermodal dependencies during the learning process poses a significant hurdle to the realization of acoustic-optical fusion.

3 | Method

In this section, we will introduce the proposed acoustic-optical fusion object detector UAMFDet. Its overall structure is shown in Figure 2. The detector is designed based on Faster-RCNN, and the backbone network for feature extraction is extended into dual branches. In the feature fusion stage, we use the proposed MDSFF module to fuse the weakly aligned acoustic and optical image features. To eliminate the differences in instance features between modalities, the MIFM module is further proposed to fuse the multi-modal instance-level features. We will introduce the MDSFF module and the MIFM module in Sections 3.1 and 3.2, respectively. In the following section, we use reference modality to represent optical modality and sensed modality to represent acoustic modality.

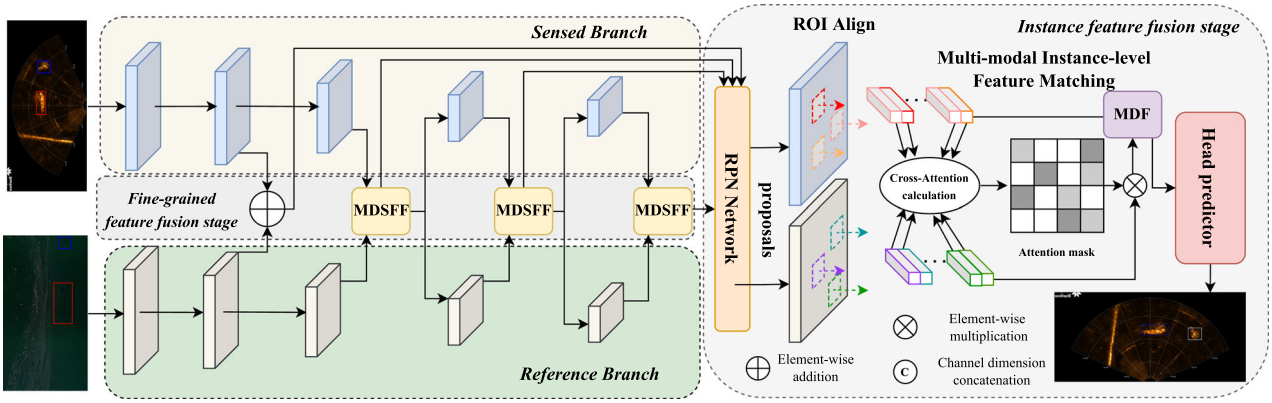


FIGURE 2 | Overrall structure of UAMFDet.

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3.1 | MDSFF Module

Given the disparities in imaging mechanisms between acoustic and optical sensors, the images they capture within a shared scene frequently exhibit substantial geometric variations and misalignment in the spatial dimension. Unlike 3D point cloud data derived from lidar, precisely determining a projection matrix to spatially align acoustic and optical images is a complex challenge. Auto-Align and similar works have harnessed the self-attention mechanism to dynamically identify feature correspondences across modalities (Chen et al. 2022). However, the substantial background noise present in images of diverse modalities can introduce redundant features, significantly compromising the efficiency of self-attention calculations when these features are incorporated. To address these issues, we introduce the MDSFF module, whose architecture is depicted in Figure 3. We compute cross-modal features for the most pertinent regions, leveraging their unique regional characteristics as a foundation. Specifically, the cross-modal regional features are fed into the positional convolution layer, where K regional positions bias $(\Delta x_k, \Delta y_k)$ are predicted to identify the regional features that are most pertinent to the specific region. This process can be formulated as follows:

$$\begin{aligned} \Delta x_k &= GELU\left[Conv_x\left(F_{\text{spec}}^{\text{sensed}} \times F_{\text{spec}}^{\text{ref}}\right)\right], k \in (0, K) \\ \Delta y_K &= GELU\left[Conv_y\left(F_{\text{spec}}^{\text{sensed}} \times F_{\text{spec}}^{\text{ref}}\right)\right], k \in (0, K) \\ x_k, y_k &= x + \Delta x_k, y + \Delta y_k \end{aligned} \tag{1}$$

where $F_{\text{spec}}^{\text{sensed}}$, $F_{\text{spec}}^{\text{ref}}$ represent sensed modal features and reference modal features at a specific location respectively. (x_k, y_k) represents coordinates of the kth relevant region. GELU is an activation function that can be expressed as $\text{GELU}(x) = 0.5x(1 + \tanh[\sqrt{\frac{2}{\pi}}(x + \lambda x^3)])$. Once the K-relevant features are obtained, the specific features of the obtained sensed modality and the candidate regional features in the reference modality are encoded. The formula can be expressed as follows:

$$\begin{aligned} F_{\text{Cand}}^{\text{ref}} &= \{F_{ref}(x_k, y_k) | k \in [1, 2, ..., K]\} \\ Q_{\text{sensed}} &= \text{FC}\left(F_{\text{spec}}^{\text{sensed}}; W_{\text{sensed}}^{Q}\right) \\ K_{\text{ref}} &= \text{FC}\left(F_{\text{cand}}^{\text{ref}}; W_{\text{ref}}^{k}\right) \\ V_{\text{ref}} &= \text{FC}\left(F_{\text{cand}}^{\text{ref}}; W_{\text{ref}}^{v}\right) \end{aligned} \tag{2}$$

where $F_{\text{Cand}}^{\text{ref}}$ represents the candidate regional features of reference modality, FV represents a fully connected layer. A multi-head self-attention module is employed to compute the attention scores between the features of the target-specific region and the K-relevant features. Subsequently, the weighted summation of these scores yields the corresponding cross-modal alignment features. The entire process can be mathematically expressed as follows:

$$\begin{aligned} W_k &= \frac{e^{Q_{\text{sensed}} \times K_{\text{ref}}^k}}{\sum_{k=1}^{K} e^{Q_{\text{sensed}} \times K_{\text{ref}}^k}} \\ F_{\text{align}}^{\text{ref}} &= \sum_{k=1}^{K} W_k F_{\text{cand}}^{\text{ref}}(k) \end{aligned} \tag{3}$$

where W_k is the calculated self-attention weight and $F_{\text{align}}^{\text{ref}}$ is the cross-modal alignment feature for the particular region.

One significant advantage of utilizing deformable attention in the self-aligned feature fusion stage is its ability to cut down the computational demands of the whole model. The traditional self-attention mechanism concentrates on establishing the dependency between specific regional features and the entire reference modality image. While this approach effectively captures the global relationship between local features and the overall image, its computational complexity scales quadratically with the image's width and height. By implementing deformable attention, we restrict the computation to the k most pertinent local features in the reference modality, corresponding to the regional features of the sensory modality. This not only establishes the dependency between corresponding features but also significantly reduces the computational complexity of the modal.

Following the alignment of cross-modal features, we have devised a multi-modal differential fusion (MDF) module to integrate complementary features from diverse modalities, thereby enriching the overall feature representation. Initially, we apply a difference operation to the features of each modality to capture their inter-modal differences. These differential features are then processed through a gated selection mechanism, which identifies and selects effective complementary features for cross-modal fusion. Specifically, we utilize global average pooling and maximum pooling to preserve maximal feature information, and these pooled features are input into a fully connected layer to predict the optimal weights for controlling

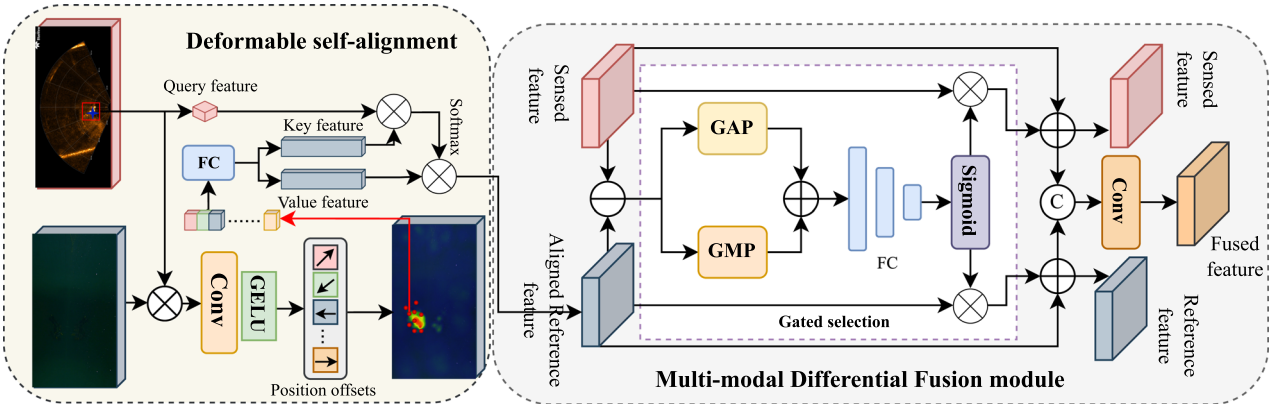


FIGURE 3 | The structure of the MDSFF module.

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feature selection. By adjusting these weights, we suppress redundant features while retaining complementary cross-modal features. The resulting complementary features are then concatenated along the channel dimension to achieve effective feature fusion. In addition, we introduce residual connections within the MDF module to prevent information loss during the fusion process and expedite the convergence speed during training.

3.2 | MIFM Module

Distinct from pixel-intensive prediction tasks, such as semantic segmentation and salient object detection, object detection also focuses on instance-level feature representation. The key to enhancing the detection performance of multi-modal object detection networks also lies in effectively fusing instance-level object features from various modalities. While features from different modalities are automatically aligned, fine-grained alignment may not precisely match the instance-level feature pairs across modalities. Simply concatenating or element-wise adding the instance-level features from two modalities can also introduce redundant noise, thereby hampering the performance of the detector. To address this issue, we design a cross-modal instance-level feature matching module that aligns instance-level features between acoustic and optical modalities. The structure of MIFM is illustrated in Figure 4. First, RPN is used to extract the potential instance-level features of N targets from sensed modality $F_{\text{Ins}}^{\text{sensed}} \in \mathbf{R}^{N\times C\times H\times W}$ and reference modality $F_{\text{Ins}}^{\text{ref}} \in \mathbf{R}^{N\times C\times H\times W}$, respectively. Then, we design a lightweight cross-modal multi-head attention mechanism to match cross-modal instance feature pairs adaptively. Specifically, global average pooling and global max pooling are used to capture the main information $F_{\text{pool}}^{\text{sensed}} \in \mathbf{R}^{N\times C}$, $F_{\text{pool}}^{\text{ref}} \in \mathbf{R}^{N\times C}$of the two modalities. The pooling features of the two models are input into the multi-head self-attention mechanism to calculate the cross-modal instance feature matching matrix $W_{N\times N}$, where W_{ij} represents the match score between the ith feature of the sensed modal and the jth feature of the reference modal. A higher matching score represents a stronger correlation between the features of the two different models. The expression is as follows:

$$\begin{aligned} F_{\text{pool}}^{\text{sensed}} &= \text{GAP}\left(F_{\text{Ins}}^{\text{sensed}}\right) + \text{GMP}\left(F_{\text{Ins}}^{\text{sensed}}\right) \\ F_{\text{pool}}^{\text{ref}} &= \text{GAP}\left(F_{pool}^{ref}\right) + \text{GMP}\left(F_{\text{pool}}^{\text{ref}}\right) \\ W_{N\times N} &= \text{Softmax}\left(F_{\text{pool}}^{\text{sensed}} \times F_{\text{pool}}^{\text{reference}^T}\right) \end{aligned} \tag{4}$$

where GAP and GMP stand for global average pooling and global max pooling, respectively, while Softmax serves as a normalized activation function. Through matrix multiplication of the acquired feature matching matrix with the modal features, we are able to extract the automatically aligned cross-modal features. Moreover, to eliminate the impact of invalid background region proposals, we leverage the feature matching matrix to quantify the similarity between each instance-level feature and other spatial location features. Following this, the Sigmoid activation function is applied to derive the normalized probability matrix M, whose expression is outlined as follows:

$$\begin{aligned} F_{\text{align}}^{\text{ref}} &= W_{N\times N} F_{\text{Ins}}^{\text{reference}} \\ M(\text{i}) &= \sigma\left(\textstyle\sum_j W(i,j)\right) = \frac{1}{1+\exp\left(-\sum_j W(i,j)\right)} \\ F_{\text{align}}^{\text{ref}'} &= M \odot F_{\text{align}}^{\text{ref}} \end{aligned} \tag{5}$$

where $M(i)$ denotes the normalized probability of match between the ith instance feature of the sensed modality and the instance feature of the reference modality. A higher value indicates a more precise cross-modal matching rate and a greater likelihood that the instance feature belongs to the target. By applying the Hadamard product of the normalized probability matrix and the features of both modalities, we can effectively suppress the instance features of the background region. Upon obtaining the aligned instance-level features, we employ the MDF module introduced in the previous section to integrate complementary instance-level features from distinct modalities. Ultimately, the fused features are fed into the detection head to

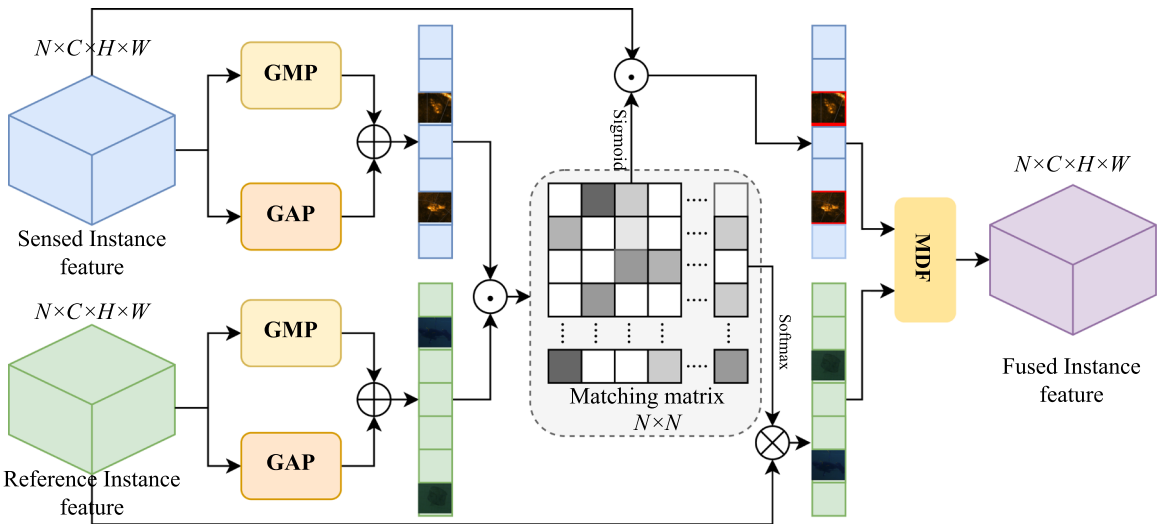


FIGURE 4 | The structure of MIFM.

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predict the target category and the coordinate regression of the detection box.

3.3 | Loss Function

Our UAMFDet is designed based on the two-stage object detection network Faster-RCNN, and its loss function is mainly composed of RPN loss, object classification loss, and box regression loss function, which can be expressed as:

$$\begin{aligned} L_{\text{total}} &= L_{\text{rpn}} + L_{cls}(p,u) + \lambda[u \geqslant 1]L_{\text{reg}}(t^u, v) \\ &= \frac{1}{N_{\text{cls}}}\sum_i L_{\text{cls}}\left(p_i^*, p_i\right) + \frac{1}{N_{\text{reg}}}\sum_i p_i^* L_{\text{reg}}\left(t_i^*, t_i\right) \\ &\quad + L_{\text{cls}}(p,u) + \lambda[u \geqslant 1]L_{\text{reg}}(t^u, v) \\ L_{\text{cls}}(p,p^*) &= -[p^* \log p + (1-p^*)\log(1-p^*)] \\ L_{\text{reg}}(t,t^*) &= \begin{cases} 0.5(t-t^*), & |t-t^*|<1 \\ |t-t^*|-0.5, & \text{otherwise} \end{cases} \end{aligned} \tag{6}$$

where p denotes probability distribution predicted by classifier, u denotes true class label, t^u true box regression parameters, v denotes corresponding predicted box regression parameters. N_{cls} represents the number of classes. N_{reg} denotes the number of anchor positions. p_i denotes the probability of positive samples. p_i^* denotes whether it is a positive sample.

4 | Experiments and Results

In this section, we will carry out a series of experiments to verify the proposed underwater multi-modal object detection network UAMFDet. First, we will introduce the proposed data set UAOF collected by self-developed underwater emergency rescue AUV. Then, we will verify the detection performance of the fusion object detection network on the UAOF data set, and compare it with the current mainstream multi-modal feature fusion methods to verify the superiority of the proposed method. In addition, we will conduct a series of ablation experiments and visualization experiments to verify the effectiveness of the proposed algorithm on underwater rescue tasks and other critical missions.

4.1 | Datasets and Evaluation Metrics

Currently, the research field lacks sets of data for underwater acoustic-optical fusion-based object detection tasks. We aim to create benchmark data sets that can serve as a data foundation to execute multi-modal object detection missions for AUVs. For data collection, we utilized a self-developed underwater emergency rescue AUV, which is equipped with an M900 forward-looking sonar and an IP Multi SeaCam at its front end. Its layout is shown in Figure 5. The M900 is a multi-beam forward-looking sonar with a range resolution of 2.5 cm and IP Multi SeaCam is an underwater optical camera with 1920 × 1080 sensor resolution. The data set was collected in Qingdao, China. After calibrating both the camera and sonar, we dispatched the AUV to capture images from various angles and distances surrounding the target of interest. Concurrently, we save the acoustic and optical images of the same scene to ensure time dimension alignment.

To enable AUVs for underwater rescue, mine exploration, and other critical missions, our data set encompasses a diverse set of 10 targets, ranging from person, mine, spherical mine, tire, ROVs with cables, small AUVs, circle fish cages, square fish cages, cone and hexagonal, as depicted in Figure 6. By eliminating redundant images with repetitive scenes and ensuring temporal synchronization between acoustic and optical images, we curated a total of 4600 valid acoustic-optical image pairs. The distribution of different target types in the data set is outlined in Table 1. Following the labeling of these 4600 image pairs, we allocated the images into training and validation sets at a ratio of 8.5:1.5, resulting in 3600 training sets and 1000 validation sets.

During the experimental process, we employed the mean average precision (mAP) index to gauge the performance of our multi-modal object detector. Specifically, we considered the average precision (AP) for each individual category, the overall

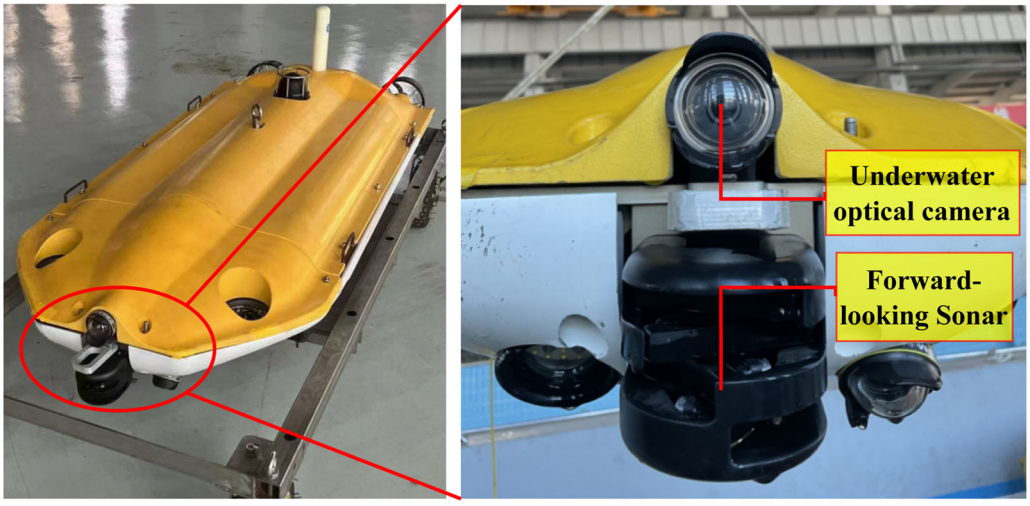


FIGURE 5 | Illustration of the underwater rescue AUV and its onboard sensors.

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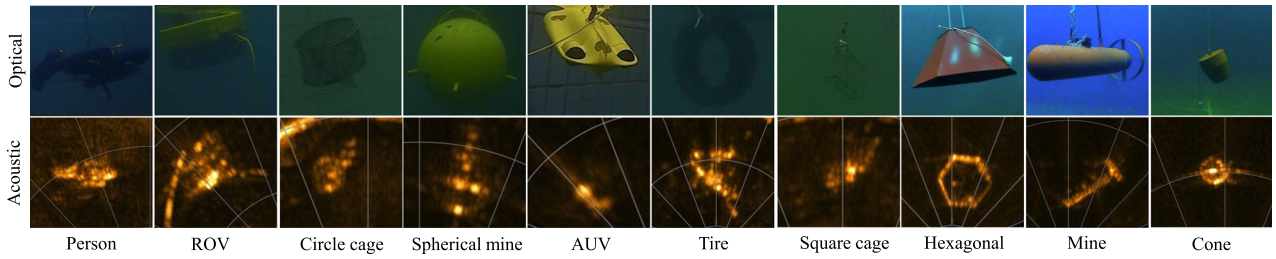


FIGURE 6 | Categories in the UAOF data set.

TABLE 1 | Category number statistics of UAOF data set.

Class	Cone	Mine	Hexagonal	Person	Circle cage	ROV	AUV	Square cage	Spherical	Tire
GT numbers	229	160	114	271	241	111	61	124	66	59

mAP for all categories, and the model complexity of the object detection network. The AP is derived from the recall–precision (P–R) curve in target detection, where recall and precision are calculated using the following equation:

$$\begin{aligned} &\text{precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \text{recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \\ &\text{AP} = \int_0^1 P(r)dr \\ &\text{mAP} = \frac{\sum_{i=1}^{c} AP_i}{c} \end{aligned} \tag{7}$$

where TP represents the correct detection as a positive sample, FP represents the wrong detection as a positive sample, FN represents the wrong detection as a negative sample, and c represents the total number of categories.

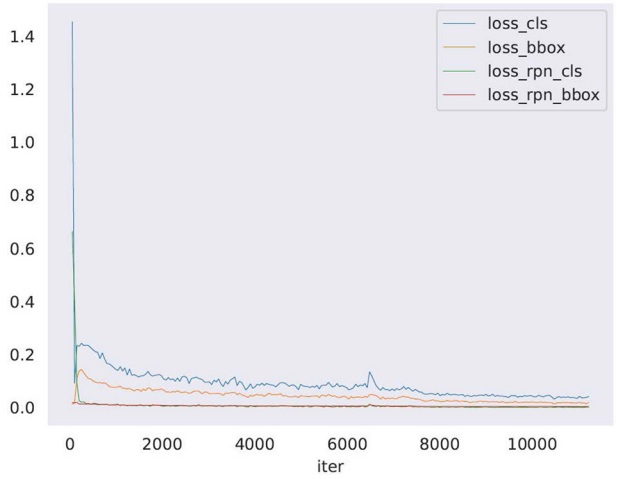


FIGURE 7 | Loss curve during the training process.

4.2 | Implement Details

We deploy our proposed multi-modal object detection network based on the MMDetection framework and train it on an Nvidia A40 card. During the training process, we initialize the network with pre-trained weights obtained from the COCO data set, setting the input image size to 640 × 640 and the batch size to 4. The training proceeds for a total of 12 epochs. To optimize the weight parameters of the network, we employ the ADAM optimizer, initiating the learning rate at 0.0001 and setting the weight decay parameter to 0.1. In addition, we configure the RPN network to retain 512 candidate boxes during the training phase and 1000 during the inference phase. As a data augmentation technique, we solely utilize random flipping during training. To ensure the fairness of our experimental comparisons, we maintain consistency in the image size, batch size, training iterations, and data augmentation methods across all the experiments. Following the training, the resulting loss function curve is depicted in Figure 7.

4.3 | Comparison of Quantitative Results

After extensive research, we discovered that there is a dearth of object detection networks tailored specifically for underwater acoustic-optical fusion object detection tasks. Consequently, in our quantitative analysis, we opted to compare against multi-modal feature fusion networks designed for other scenarios. For a fair comparison of feature fusion methods, we disregarded their decoders and solely applied their proposed multi-modal feature fusion techniques to object detection tasks during our comparative experiments. Specifically, we selected CMX (Zhang, Liu, et al. 2023), SA-GATE, ACNet, and RedNet (Jiang et al. 2018) for RGB-D multi-modal tasks, as well as EAEFNet, GMNet, SFAF-MA (He et al. 2023), and CAINet (Lv, Liu, and Li 2024) for RGB-T tasks, to fuse the features of acoustic and optical images. CMX realizes feature complementarity between modalities by modifying them mutually in both spatial and channel dimensions. SA-Gate employs a separation and aggregation gating mechanism to filter out invalid features, thereby achieving feature calibration across modalities. ACNet leverages an attention mechanism to activate effective features between modalities. EAEFNet utilizes explicit attention to modulate features of diverse modalities, enabling efficient fusion between state modalities. GMNet achieves effective hierarchical fusion through three stages: primary, intermediate, and advanced feature fusion, establishing a complementary relationship between multi-modal features and long-term

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context in spatial and channel dimensions. Similarly, CAINet also achieves an effective fusion of multi-modal features by establishing this complementary relationship. SFAF-MA generates adaptive weights through an attention mechanism to fuse features of different modalities.

To ensure a fair comparison, all multi-modal fusion methods are implemented on the unified baseline detection network Faster-RCNN. This network comprises two backbone networks for extracting features from different modalities, an RPN network for generating instance candidates, and a detection head for refining detection results. The quantitative comparison results are presented in Table 2.

As evident from the table, the multi-modal fusion methods exemplified by CMX and GMNet, designed to fuse RGB-D and RGB-T modes, respectively, rely on aligned images for feature integration across modalities. While these methods offer an improvement over the baseline faster R-CNN in fusing underwater acoustic and optical modal features, the inherent heterogeneity and significant misalignment of acoustic-visual images hinder capturing the positional dependencies between modalities during fusion, introducing redundant features. For instance, RedNet lags behind the baseline due to the additional noise introduced by the fusion of highly misaligned features using a simplistic method. In contrast, our UAMFDet framework automatically identifies feature dependencies between modalities through an MDSFF module. It then integrates fine-grained features post-alignment and matches acoustic and optical instance features using the MIFM module. Consequently, our method outperforms most categories in detection performance, surpassing the baseline method by 3.5% in mAP index. Furthermore, it improves the average detection accuracy of GMNet, which ranks second, by 1.9%.

To validate the robustness of our fusion method, we investigate its adaptability to a one-stage object detection network. We select YOLOX from the YOLO series as our baseline, expanding its backbone network to a dual-branch architecture to independently extract features from distinct modalities. Subsequently, we implement our proposed feature fusion approach on this baseline. Given that YOLOX lacks an RPN network for candidate box extraction, we forgo the MIFM module for matching and fusing intermediate-level features. So we focus on evaluating the effectiveness of our fine-grained fusion method MDSFF within the one-stage detector. The results of this experiment are presented in Table 3. The MDSFF module, as compared to simple element-wise addition and other fusion methods like EAEFNet and CAINet, exhibits superior fusion capabilities for optical and acoustic modaliies within the one-stage object detector network, surpassing the baseline YOLOX by 1.1% in mAP index. Notably, the performance gains of alternative object detection approaches are relatively modest. When evaluating the efficiency of models, we considered both the number of model parameters and computational complexity. As indicated in the table, our approach introduces an increment of 12.9 M parameters and 13.1 GFLOPs compared to the YOLOX baseline. However, it's worth noting that the actual parameter count and computational complexity of the MDSFF module are lower than these figures, as we also attribute the growth to the additional backbone network required for

TABLE 3 | Analysis of model parameters and complexity with one-stage detectors.

Detector	Fusion method	mAP (%)	Params (M)	GFLOPs
YOLOX	—	83.3	8.9	13.3
	Eltwise. Sum	83.5	13.2	18.8
	EAEFNet	83.3	14.9	19.1
	CAINet	83.6	13.5	19.2
	ours	84.4	21.8	26.4

TABLE 2 | Comparison results of different methods on UAOF data set.

	AP (%)										
Method	Cone	Mine	Hexagonal	Person	Circle cage	ROV	AUV	Squre cage	Spherical	Tire	mAP (%)
Faster-RCNN	72.9	73.3	77.9	90.4	90.8	91.1	74.6	77.6	90.1	89.8	82.8
CMX	75.4	75.6	78.1	90.8	90.5	90.9	74.5	79.8	90.3	90.6	83.6
SA-Gate	75.5	80	77.8	90.5	90.7	90.8	71.3	80.8	89.1	89.3	83.6
ACNet	76.1	78.8	78.3	90.6	90.7	90.5	72.7	79.1	89.2	89.9	83.4
RedNet	75.5	77.1	77.2	90.4	90.4	90.9	66.4	80.2	89.6	89.0	82.7
EAEFNet	74.1	78.5	77.7	**90.7**	90.6	91.2	68.5	79.1	89.5	90.4	83.0
GMNet	75.7	80.1	78.1	90.5	**90.9**	90.9	**76.4**	80.6	89.3	**91.1**	84.4
CAINet	76	81.8	78.3	90.2	90.8	90.9	73.2	79.8	**90.5**	90.6	84.2
SFAF-MA	75.6	79.2	78.5	**90.7**	90.5	90.3	74.1	79.6	90.3	90.1	83.9
Ours (UAFMDet)	**78.8**	**82.3**	**78.8**	90.6	90.8	**98.4**	76	**85.4**	90.3	90.9	**86.3**

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extracting features from the reference modality in multi-modal tasks.

We further visualize the mAP change curve and confusion matrix during the training process of the aforementioned method, presented in Figures 8 and 9, respectively. As depicted in Figure 8, the mAP exhibits a consistent upward trajectory with an increasing number of training rounds, indicating that our UAFMDet achieves state-of-the-art detection results compared to other methods. The confusion matrix is a crucial tool for assessing model performance in object detection networks, which aids in evaluating the efficacy of the model. It provides insights into key metrics, such as the model's accuracy in recognizing different object classes. Each row of the confusion matrix corresponds to the true class, while each column represents the predicted class. The values within each cell indicate the frequency of true classes being predicted as such. Figure 9 reveals that UAFMDet predicts the true target category more accurately than other feature fusion methods.

4.4 | Ablation Experiment

To further verify the effectiveness of the MDSFF module and the instance-level feature matching module (MIFM), we conduct ablation experiments on the UAOF data set. In the experimental stage, Faster-RCNN is used as the baseline method, and the performance performance of each module is shown in Table 4.

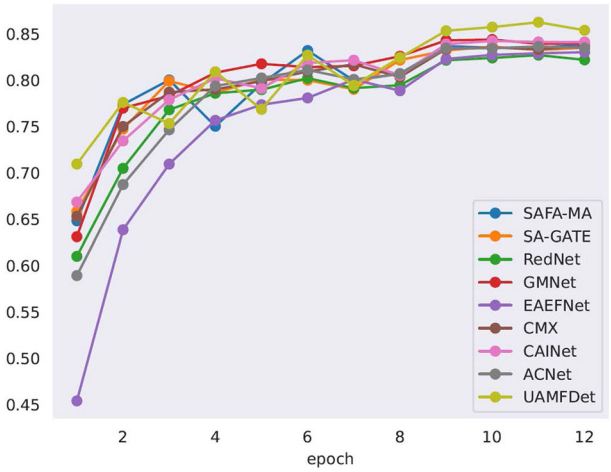


FIGURE 8 | The mAP curve of different methods.

"Eltwise. Sum" in the table represents the simple addition of the features of different modalities to complete the fusion of acoustic-optical features. From the results of ablation experiments, it can be seen that the simple feature fusion method of element-wise addition cannot effectively fuse highly misaligned acoustic-optical modalities, and even introduces additional feature noise. Compared with the baseline method Faster-RCNN, the detection accuracy is even decreased by 0.1%. After using the MDSFF module to automatically align the fine-grained features, the acoustic-optical multi-modal features are effectively fused, and the detection accuracy is improved by 2.4% compared with the baseline method Faster-RCNN. After further using the MIFM module to match the instance-level feature pairs and fuse them, the detection accuracy is further improved by 1.1%. Through the ablation experiments described above, we verified the effectiveness of the proposed MDSFF and MIFM modules.

4.5 | Visualized Analysis

We randomly sampled test images from the UAOF data set and visualized the detection outcomes of CMX, EAEFNet, GMNet, SFAF-MA, and UAMFDet, as depicted in Figure 10. The first and second rows showcase the ground truth boxes in the optical and acoustic modal, respectively. Upon comparison, we observed that detection networks utilizing CMX and GMNet feature fusion methods tend to repeatedly detect the target object. For instance, in the first column, CMX accurately detects the hexagonal but incorrectly labels it as a cone. In the second column, GMNet initially detects the ROV target accurately but subsequently misidentifies it as a personal target. Drowning people is the core target of underwater emergency rescue.

TABLE 4 | Analysis of model parameters and complexity.

Method	Eltwise. Sum	MDSFF	MIFM	mAP (%)
Faster-RCNN				82.8
	✓			82.7
		✓		85.2
UAMFADet		✓	✓	86.3

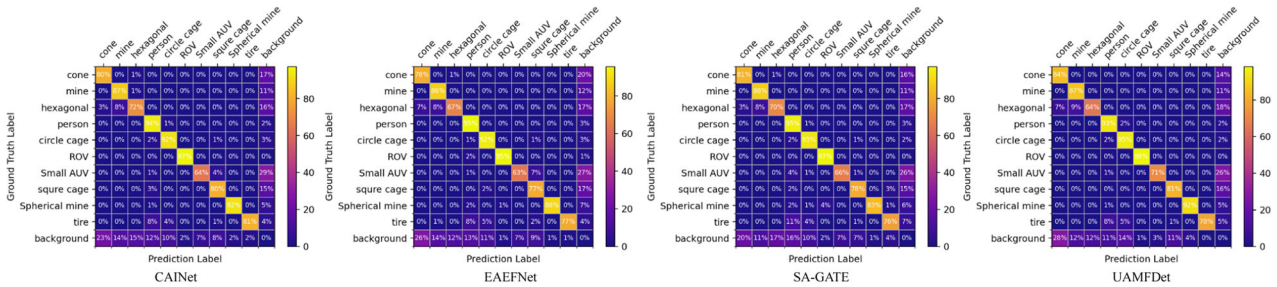


FIGURE 9 | The confusion matrix of different methods.

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EAEFNet and SFAF-MA exhibit missed detections, like failing to recognize the person target in the last column. But our UAMFDet made the right decision, which proves its promising application in underwater emergency rescue missions. In addition, CMX and SFAF-MA exhibit false detections, such as CMX mistaking noise in the image for a target in the third column. In contrast, UAMFDet avoids such issues during detection, delivering more satisfactory results.

To validate that UDMFDet effectively establishes feature dependencies between highly misaligned acoustic-optical modalities, we selectively visualize the attention masks that generate for the entire reference modal at specific sensed modal locations, as illustrated in Figure 11. Specifically, we present the visualization of the attention masks corresponding to four points: P1, P2, P3, and P4. For P1 and P2, the model accurately highlights the region in the

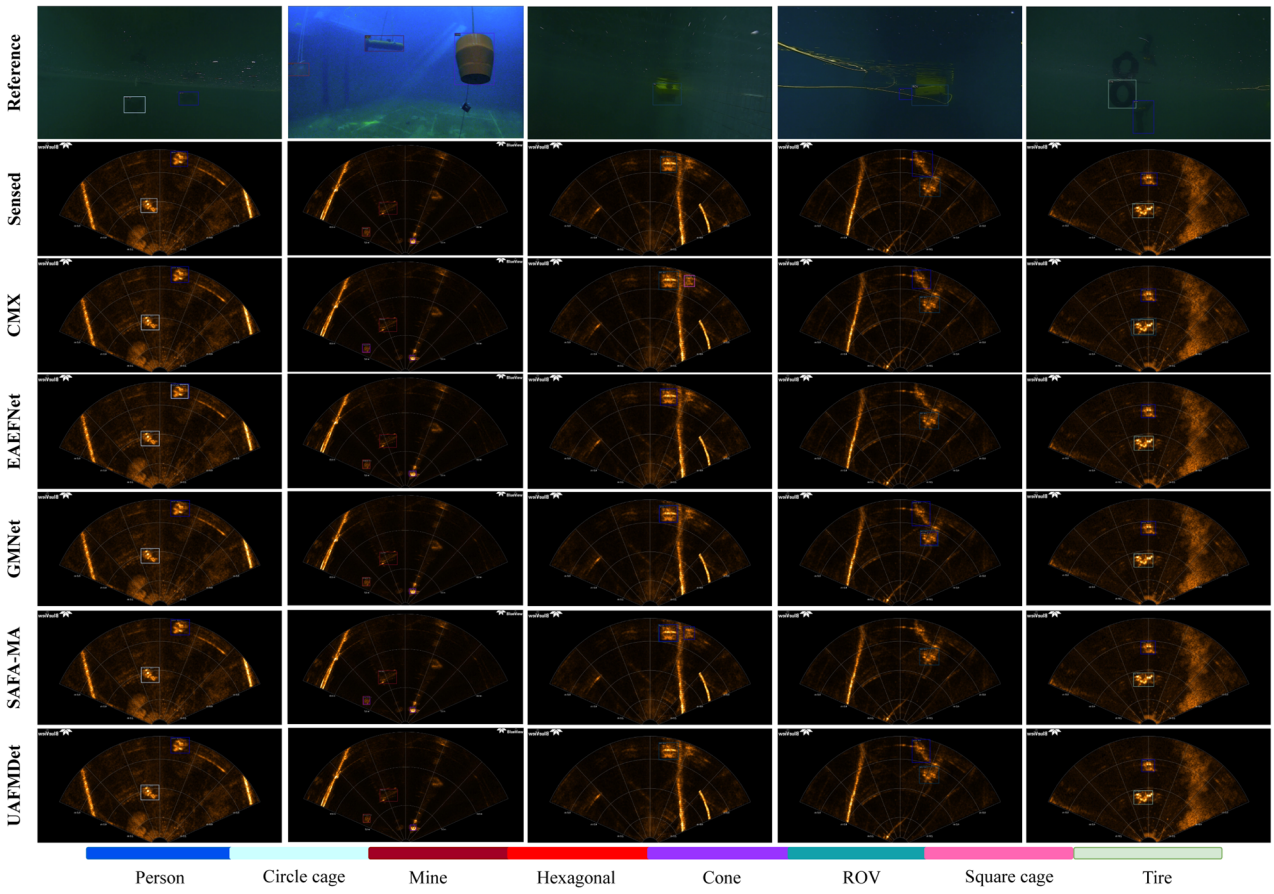


FIGURE 10 | Detection results of different methods on UAOF datasets.

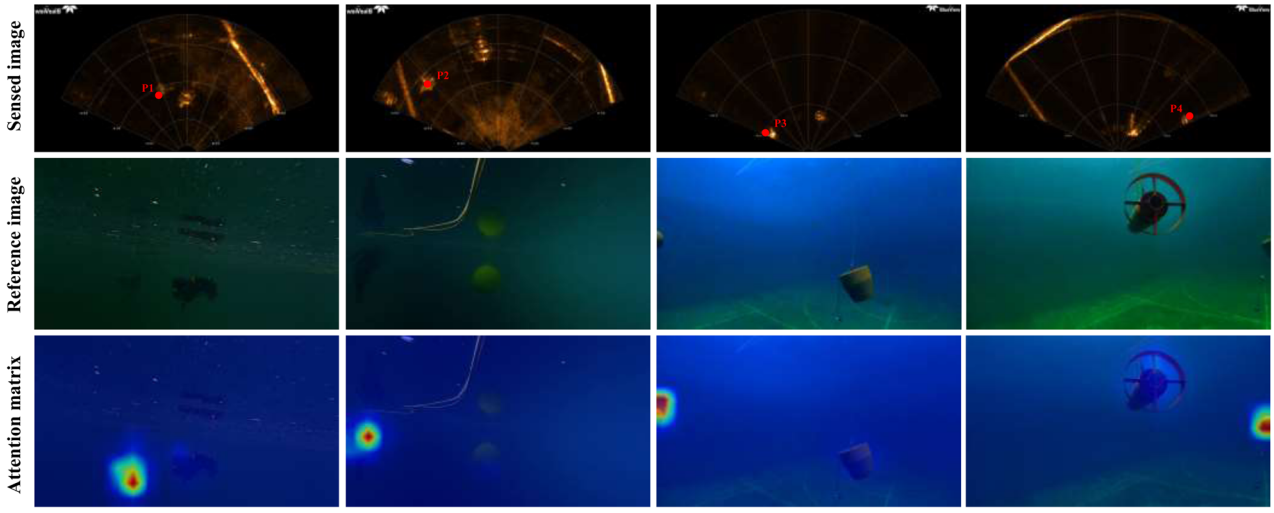


FIGURE 11 | Visualization attention matrix of a specific point.

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reference modal that contains semantically equivalent information to the original position of the sensed modal. This demonstrates that despite significant spatial misalignment between models, our UAMFDet can adaptively capture cross-modal target feature dependencies. Notably, even when targets are severely occluded or positioned at the edge, as evident in the attention masks for P3 and P4, this relationship remains robust. The visualization results underscore the capability of our feature fusion approach to capture long-range, semantically consistent deep features.

4.6 | Effeplexity with one-stage detect

To further validate the detection efficacy of UAMFDet on land-based datasets, we conducted experiments using the RGB-T object detection data set M3FD (Figure 12). This data set comprises six target categories: car, bus, lamp, people, truck, and motorcycle. Following the application of the calibration matrix, the visible light and infrared multispectral images in the data set exhibited high spatial alignment. The experimental outcomes are summarized in Table 5. As evident from the table, our UAMFDet outperforms the baseline Faster-RCNN method, achieving a 2.1% increase in average detection accuracy. Even when compared to EAEFNet, a method tailored for RGB-T modal feature fusion, our UAMFDet still manages to enhance the average detection accuracy by 0.2%. This underscores the positive role played by UAMFDet in fusing aligned multi-modal land-based images.

4.7 | Discussion

In this paper, we explore a novel multi-modal object detection task called underwater acoustic-optical fusion object detection. We introduce UAMFNet, a framework that enhances the detection performance of underwater object detection networks by fusing highly misaligned inter-modal features. Compared to other feature fusion methods reliant on modality alignment, UAMFNet exhibits superior detection performance on the UAOF data set. However, as this work is a preliminary investigation into underwater multi-modal object detection, UAMFNet still faces certain limitations. While the deformable attention mechanism in MDSFF reduces the computational burden of cross-modal global modeling, predicting the most pertinent region in the reference modality for each sensed modality region remains computationally intensive. Furthermore, the UAOF data set was created under limited underwater imaging conditions, whereas in real-world applications, underwater vehicles encounter diverse waters and operational scenarios, necessitating a more comprehensive data set containing a wider range of scenes.

In our future work, we aim to further refine our acoustical-optical feature fusion module, striving for a more lightweight design that can be seamlessly integrated into low-power edge computing devices on AUVs. In addition, we plan to gather more acoustic-optical paired images from various underwater environments to enrich the diversity of the UAOF data set, ultimately enhancing the ability of AUV to perform target detection across a broad range of operational conditions.

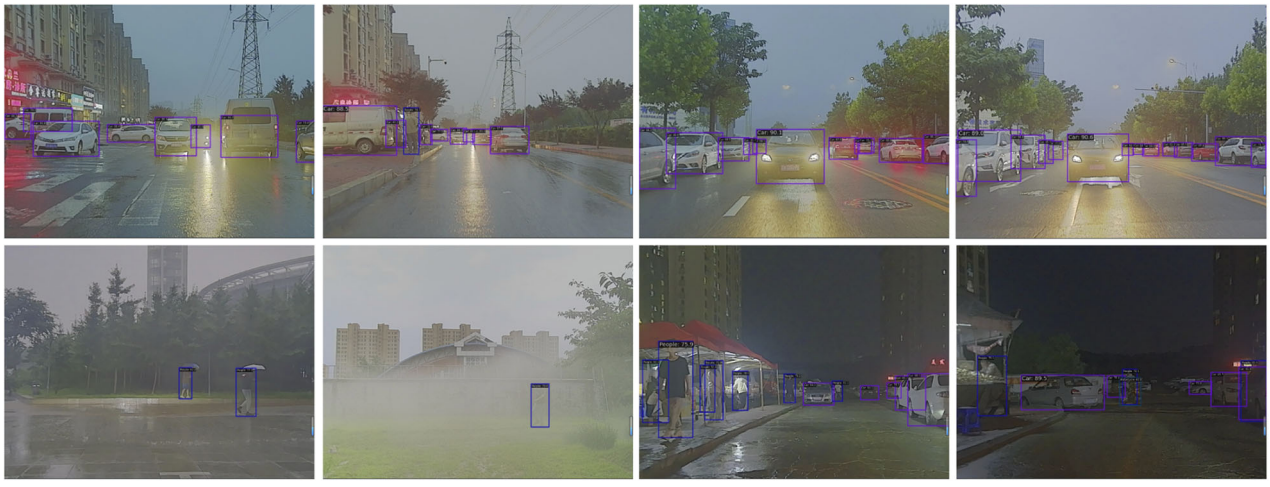


FIGURE 12 | Detection results of different methods on M3FD datasets.

TABLE 5 | Comparison results on M3FD data set.

Method	Car (%)	Bus (%)	Lamp (%)	People (%)	Truck (%)	Motorcycle (%)	mAP (%)
Faster-RCNN	69.9	69.8	9.1	36.7	68.0	52.8	51.1
Faster-RCNN w EAEFNet	68.1	78.5	9.1	36	72.4	53.8	53.0
UAMFDet	68.6	75.6	9.5	35.8	73.3	57.0	53.0

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5 | Conclusion

This paper delves into a novel underwater acoustic-optical fusion-based object detection task and introduces UAMFDet, a multi-modal object detection network that leverages the fusion of acoustic and optical modalities. To address the challenge of misalignment between acoustic and optical modal images in the spatial dimension, UAMFDet fuses multi-modal features at two distinct levels: fine-grained and instance-level. During the feature extraction phase, we introduce an MDSFF module. This module establishes feature dependencies across modalities through a deformable attention mechanism, enabling differential fusion of self-aligned multi-modal features for fine-grained modality-specific feature integration. Furthermore, we extract instance-level features via an RPN and present a MIFM module. This module pairs instance-level features of from different modalities, followed by differential fusion to achieve complementary instance-level feature integration. Through these two-tiered fusion processes, UAMFDet effectively aligns and integrates acoustic and optical features at varying degrees of fine-grained precision and instance-specificity. In addition, we establish the first benchmark data set, UAOF, for acoustic-optical fusion object detection and conduct extensive experiments on UAOF to validate the effectiveness of UAMFDet.

Acknowledgments

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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